

PAPER_199: F_UBii Taxonomy Part 2 — Cosmological and Dark Sector Buoyancy Forces

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$\rho_{\Lambda}^{\text{UQFF}} = \rho_{\Lambda}^{\text{obs}} \cdot \left(1 + \kappa^2 \cdot [SSq]^2\right) = \rho_{\Lambda}^{\text{obs}} \times 1.0000000812$$

Abstract

This paper catalogs the second major group of F_UBii variants from the BB_C_Equations_04Sept2026 (177-page equation catalogue): cosmological and dark sector buoyancy forces. Covered are anyon systems, dark energy, inflation, gravitational waves, Loop Quantum Cosmology (LQC) bounce and perturbation spectra, Bekenstein-Hawking entropy, Hawking evaporation lifetime, reheating, Big Bang Nucleosynthesis, reionization, baryon-photon ratio, convective turnover time, CMB angular power spectrum, recombination, NFW dark matter profiles, SIDM core formation, void density evolution, and peculiar velocity. Each F_UBii,X form uses the universal F_rel/E_LEP scaling with a system-specific physical expression F_X.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Dark Sector and Quantum Gravity Variants

1.1 Dark Energy Buoyancy

$$F_{\text{UBii,DE}} = -F_{\text{rel}} \times (\kappa_{\text{DE}} \cdot c^2 / E_{\text{LEP}}) \times Q_{\text{wave}}(z) \times (8\rho_{\text{G}}^{\text{tot}}/3) \times (1+w(z))$$

$$\kappa_{\text{DE}}(a) = \kappa_{\text{DE},0} \cdot \exp(3 \int_0^a (1+w(a'))/a' da')$$

$$w(a) = w_0 + w_a(1-a) \quad (\text{Chevallier-Polarski-Linder parametrization})$$

Source: BB_C_Equations item 721

1.2 Inflation Buoyancy

$$F_{\text{UBii,inf}} = F_{\text{rel}} \times (V(\phi) / E_{\text{LEP}}) \times Q_{\text{wave}} \times 3H^2 \times e^N/(1+e)$$

$V(\phi)$ = inflaton potential

N = number of e-folds

e = slow-roll parameter: $e = (\ddot{\phi}/H \cdot M_{\text{pl}})^2/2$

Source: BB_C_Equations item 724

1.3 Gravitational Wave Energy Density Buoyancy

$$F_{UBii,GW} = -F_{rel} \times (\dot{\rho}_{GW} / E_{LEP}) \times Q_{wave} \times (32\pi G \dot{\rho}^2 / c^2) \times e^{-t/t}$$

$\dot{\rho}_{GW}$ = gravitational wave energy density

$\dot{\rho}$ = time derivative of strain

Source: BB_C_Equations item 727

1.4 Anyon System Buoyancy

$$F_{UBii,anyons} = -F_{rel} \times (E_{anyons} / E_{LEP}) \times Q_{wave} \times g(r,t) \times \exp(-d_c^2 / (2s^2))$$

E_{anyons} = anyon system energy (2D topological)

$g(r,t)$ = UQFF gravitational field at position

Gaussian factor from density fluctuation s

Source: BB_C_Equations item 710

2. Loop Quantum Cosmology (LQC) Variants

2.1 LQC Perturbation Spectrum Buoyancy

$$F_{UBii,lqcp} = -F_{rel} \times (P(k) \dot{\rho}^{n_s-1} \cdot (1+k/k_*)^{-a} / E_{LEP}) \times Q_{wave}$$

k_* = bounce scale (LQC pre-bounce phase)

a = UV suppression exponent

Modifies primordial power spectrum at Planck-scale modes

Source: BB_C_Equations item 1431, 1254

2.2 LQC Effective Friedmann Buoyancy

$$F_{UBii,lqcf} = F_{rel} \times (H^2 = 8\pi G \dot{\rho} / 3 \cdot (1 - \dot{\rho} / \dot{\rho}_{crit}) / E_{LEP}) \times Q_{wave}$$

$\dot{\rho}_{crit} = 0.41 \cdot \dot{\rho}_{Planck} \sim 10^{13} \text{ g/cm}^3$

Bounce condition: $H=0$ at $\dot{\rho}=\dot{\rho}_{crit}$ (avoids singularity)

Source: BB_C_Equations item 1252, 1604

2.3 LQC Bounce Timescale Buoyancy

$$F_{UBii,bounc} = F_{rel} \times (t_b \sim v(3/(8\pi G \dot{\rho}_{crit})) \sim t_{Pl} \sim 10^{-43} \text{ s} / E_{LEP}) \\ \times Q_{wave} \times [H \sim 0 \text{ at bounce, duration} \sim 1/\dot{\rho}_{bounce}]$$

Source: BB_C_Equations item 1257, post-1608

3. Black Hole Thermodynamics Variants

3.1 Bekenstein-Hawking Entropy Buoyancy

$$F_{UBii,bhent} = F_{rel} \times (S = k_B \cdot c^3 \cdot A / (4Gh) = 4\pi \cdot k_B \cdot GM^2 / (hc) / E_{LEP}) \times Q_{wave}$$

$A = 4\pi r_s^2$ (Schwarzschild horizon area)

$r_s = 2GM/c^2$

Holographic: $S \sim A/l_{Pl}^2$ (area law)

Source: BB_C_Equations item 1092, 1248

3.2 Evaporation Lifetime Buoyancy

$$F_{UBii, \text{evap}} = -F_{\text{rel}} \times (t_{\text{evap}} = 5120 \text{pG}^2 \text{M}^3 / (\text{hc}^4) / E_{\text{LEP}}) \times Q_{\text{wave}} \times [P = sAT^4 \sim \text{hc}^2 / \text{M}^2]$$

Power $dM/dt = -P/c^2$ (mass loss rate)

Source: BB_C_Equations item 1601, 1250

4. Cosmological Nucleosynthesis and Reionization Variants

4.1 Big Bang Nucleosynthesis Deuterium Bottleneck

$$F_{UBii, \text{deb}} = -F_{\text{rel}} \times (t_D = v(3/(32 \text{pG}^2_{\text{rad}})) \sim 180 \text{ s } (T \sim 0.1 \text{ MeV}) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$\rho_{\text{rad}} = p^2 k T^4 / (30 h^3 c^5) \cdot g_*$ (radiation density)

g_* = effective degrees of freedom (~ 10 at D formation)

Weak freeze-out at $T \sim 1 \text{ MeV}$; photodissociation until $T < 0.1 \text{ MeV}$

Source: BB_C_Equations item 1809, 1536

4.2 Baryon-to-Photon Ratio Buoyancy

$$F_{UBii, \text{eta}} = F_{\text{rel}} \times (\eta = n_b/n_\gamma = 6 \times 10^{-10} / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\text{Freeze-out: R...}]$$

$n_\gamma = 410 \text{ cm}^{-3}$ (CMB photon density today)

Fit from D, ^3He , ^4He , ^7Li abundances

Source: BB_C_Equations item 1701, 1534

4.3 Reionization Front Buoyancy

$$F_{UBii, \text{reion}} = F_{\text{rel}} \times (d\eta_e/dt = \eta_\gamma \cdot e_{\text{esc}} \cdot f_* - a_B \cdot n_e^2 \cdot C / E_{\text{LEP}}) \times Q_{\text{wave}}$$

x_e = ionized fraction, $e_{\text{esc}} \sim 0.1-0.3$ (escape fraction)

C = clumping factor

Source: BB_C_Equations item 1684

5. CMB and Recombination Variants

5.1 CMB Angular Power Spectrum Buoyancy

$$F_{UBii, \text{cmb}} = F_{\text{rel}} \times (C_l = 2/p \cdot \int k^2 dk \cdot P(k) \cdot |\Delta_l^T(k)|^2 / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$P(k) \propto k^{n_s-4}$ (primordial power, $n_s \sim 0.965$)

Transfer Δ_l^T : Sachs-Wolfe (large scales) + acoustic (small scales)

Source: BB_C_Equations item 1310, 1080

5.2 Recombination Optical Depth Buoyancy

$$F_{UBii, \text{recomb}} = -F_{\text{rel}} \times (t(z) = \int_{z_{\text{re}}}^8 n_e(z') \cdot s_T \cdot c \cdot (dt/dz') dz' / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$z_{\text{re}} \sim 7.7$ (reionization redshift, Planck 2018)

s_T = Thomson cross-section ($6.65 \times 10^{-29} \text{ m}^2$)
Source: BB_C_Equations item 1313

6. Dark Matter Variants

6.1 NFW Density Profile Buoyancy

$$F_{UBii,nfw} = -F_{rel} \times \left(\rho(r) = \rho_s / ((r/r_s) \cdot (1+r/r_s)^2) \right) / E_{LEP} \times Q_{wave}$$

ρ_s = characteristic density, r_s = scale radius
Universal CDM halo form
Source: BB_C_Equations item 1326

6.2 NFW Rotation Curve Buoyancy

$$F_{UBii,nfwrot} = F_{rel} \times (v(r)^2 = 4\pi G \rho_s \cdot r_s^2 \cdot [\ln(1+x) - x/(1+x)]/r) / E_{LEP} \times Q_{wave} \times x=r/r_s$$

Flat for $r \gg r_s$
Source: BB_C_Equations item 1535

6.3 SIDM Core Formation Buoyancy

$$F_{UBii,sidm} = -F_{rel} \times (G = \rho \cdot s \cdot v/m) / E_{LEP} \times Q_{wave} \times \ln(0.02N)$$

$t_{core} \sim (\rho \cdot s/m)^{-1} \sim 10^{10} \cdot (\rho/108 \text{ M}_\odot/\text{kpc}^3)^{-1} \cdot (s/m/1 \text{ cm}^2/\text{g})^{-1} \text{ yr}$
Core forms when $G \cdot t \sim 1$
Source: BB_C_Equations item 1249, 1264

7. Void and Peculiar Velocity Variants

7.1 Void Density Evolution Buoyancy

$$F_{UBii,voidden} = -F_{rel} \times (d_v(a) = -3/5 \cdot (0_m \cdot a + 0_v)^{-3/2} \cdot d_v0) / E_{LEP} \times Q_{wave}$$

a = scale factor, d_v0 = initial underdensity
Integration: $d \rho / a^{-1}$ in matter domination inside void
Source: BB_C_Equations item 1940, 1338

7.2 Peculiar Velocity Buoyancy

$$F_{UBii,pec} = F_{rel} \times (v_{pec} = -(fH/3) \cdot \rho d(r) \cdot r \cdot dr/r^2) / E_{LEP} \times Q_{wave}$$

$f \sim 0_m^{\{0.55\}}$ (growth rate)
Spherical void: integrate Poisson equation
Source: BB_C_Equations item 1341

8. Convection and Dynamo Variants

8.1 Convective Turnover Time Buoyancy

$$F_{UBii,conv} = F_{rel} \times (t_{conv} = H_p/v_{conv} ; v_{conv} \sim (a^2 g dT \cdot H_p / (4T))^{1/3} / E_{LEP}) \times Q_{wave}$$

H_p = pressure scale height
 $a \sim 2$ (mixing length parameter)
Source: BB_C_Equations item 1533

8.2 Magnetic Field Reversal Buoyancy (Dynamo Parity)

$$F_{UBii,rev} = F_{rel} \times (l_{rev} = (a_{dynamo} \cdot ?)^{1/2} \cdot l_{force} / E_{LEP}) \times Q_{wave}$$

a_{dynamo} = dynamo a-coefficient ($\sim v/3$)
 $?$ = magnetic resistivity
Growth vs diffusion sets reversal scale
Source: BB_C_Equations item 1863, 1238

8.3 Kazantsev Dynamo Buoyancy

$$F_{UBii,dyn} = F_{rel} \times (dE_M/dt = (3/2) \cdot E_M/t_{eddy} \cdot (Re_m^{1/2} - 1) / E_{LEP}) \times Q_{wave}$$

$Re_m \sim 10^{10}$ (magnetic Reynolds number in galaxy clusters)
 $t_{eddy} = l/v \sim \text{Myr}$ (eddy turnover for $l \sim \text{kpc}$)
Source: BB_C_Equations item 1234, 1411

9. Star Formation and Feedback Variants

9.1 Metal Enrichment Buoyancy

$$F_{UBii,metal} = F_{rel} \times (Z = Y \cdot \text{SFR} / ?_{gas} - Z \cdot ?_{out} / M_{gas} / E_{LEP}) \times Q_{wave} \times (Z \sim 0.1)$$

$Y \sim 0.02$ (stellar yield)
Steady state: $Z = Y \cdot \text{SFR} / ?_{out}$
Source: BB_C_Equations item 1395, 1571

9.2 Photoevaporation (Exoplanet) Buoyancy

$$F_{UBii,photo} = F_{rel} \times (?_{evap} = e \cdot L_X \cdot R_p^3 / (GM_p^2 \cdot K(?)) / E_{LEP}) \times Q_{wave}$$

$L_X \sim 10^{27} \text{ erg/s}$ (host star X-ray luminosity)
 $K(?)$ = penetration correction factor
Source: BB_C_Equations item 1496, 1490

10. References

- grok_share_7514fe.txt lines 2766-2900, 6000-6380 (BB_C_Equations_04Sept2025.pdf Part 2 catalogue)
- PAPER_198: F_UBii Taxonomy Part 1 (Compact Objects)
- PAPER_196: Triadic Master Equation System